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Abstract

This report presents our solution for the VAND4.0 Industrial Track on MVTec AD 2 under the regular setting. The task requires pixel-level anomaly segmentation using only defect-free training images, while maintaining a class-agnostic pipeline across diverse industrial categories. We build our method upon a modified INP-Former backbone and introduce class-agnostic random-noise perturbations as synthetic abnormal counterexamples. Instead of treating these perturbations as realistic defect templates, we use them to explicitly improve normal–abnormal separability through an asymmetric counterexample objective. A lightweight residual localization branch is further incorporated to enhance dense anomaly prediction. The entire pipeline uses a shared architecture, unified training strategy, and global validation-derived threshold across all categories. Experimental results on MVTec AD 2 show that the proposed method achieves strong performance under the unified regular setting.

1. Introduction

1.1. Background

Industrial anomaly detection aims to localize visual patterns that deviate from normal products, and is a key component of automated quality inspection. In real production lines, anomalous samples are rare, diverse, and often unavailable before deployment. Exhaustively collecting pixel-level annotations for all possible defects is therefore impractical. This makes unsupervised anomaly detection a natural formulation, where models are trained primarily on defect-free images and are expected to discover unseen abnormal patterns at test time.

MVTec AD 2 further pushes this setting toward realistic industrial inspection. Compared with conventional anomaly benchmarks, it introduces more challenging visual conditions, including strong illumination variations, transparent or overlapping objects, large intra-class normal variations, and extremely small defects. These factors make anomaly localization difficult in two aspects. First, normal appearance changes may produce high anomaly responses, leading to false positives. Second, subtle defects may be suppressed by feature abstraction or score smoothing, leading to missed detections. Thus, the task requires not only discriminative anomaly scoring, but also stable pixel-level localization.

A practical solution should also scale across product categories. Category-specific models or manually tuned post-processing rules are costly to maintain and often fragile under distribution shifts. Unified anomaly detection addresses this issue by using a single model and a shared inference pipeline across categories. However, this also imposes a stronger requirement: the model must separate normal and abnormal responses under diverse textures, structures, defect scales, and imaging conditions without relying on category-specific assumptions.

1.2. Challenge Description

This report targets the Industrial Track of the VAND4.0 Challenge under the regular setting. The benchmark is MVTec AD 2, and the objective is pixel-level anomaly segmentation. For each test image, the model is required to produce a continuous anomaly map and a thresholded binary mask. The official evaluation focuses on segmentation quality, with SegF1 as the primary metric.

The challenge setting introduces three central difficulties. First, only defect-free real images are available for training, while test-time defects are open-set and cannot be enumerated in advance. The model must therefore learn a general criterion of abnormality rather than a fixed set of defect appearances. Second, the method must be class-agnostic. The same architecture, training strategy, inference procedure, thresholding rule, and post-processing pipeline should be applied to all categories. Third, threshold generalization is critical. Since the final prediction is a binary mask, unstable anomaly-score calibration across categories or lighting conditions can directly degrade SegF1.

To address these requirements, we develop a unified unsupervised anomaly segmentation framework. The method is built upon a modified INP-Former backbone and uses class-agnostic random-noise perturbations to construct synthetic abnormal counterexamples. Rather than treating these perturbations as realistic defect templates, we use them to explicitly enlarge the separation between normal and abnormal responses through an asymmetric divergence objective. A residual localization module is further introduced to strengthen dense pixel-level anomaly prediction. The resulting pipeline is unified, reproducible, and avoids category-specific tuning, making it well aligned with the regular industrial setting of the challenge.

2. Methodology

2.1. Model Design

Our method follows a unified unsupervised anomaly detection paradigm. Given only defect-free training images, we construct synthetic abnormal regions using a class-agnostic random-noise perturbation strategy and train a single model across all categories. The framework is built upon a modified INP-Former backbone, where intrinsic normal representations are extracted from each input image and used to reconstruct normal features. To improve normal–abnormal separability, we remove the gather loss in the baseline implementation and introduce an asymmetric divergence objective that treats synthetic abnormal regions as counterexamples rather than restoration targets. A lightweight residual localization module is further added to strengthen dense pixel-level anomaly prediction.

The overall pipeline contains three main components: 1) a modified INP-Former backbone for unified normality modeling; 2) an asymmetric counterexample objective for enlarging the response gap between normal and abnormal regions; and 3) a residual localization branch for improving fine-grained anomaly maps. All components are shared across categories, and no category-specific model, threshold, or post-processing rule is introduced.

2.1.1. Approach

The challenge requires a model that can localize unseen defects without using real anomalous training samples. A pure reconstruction-based objective is insufficient in this setting because a high-capacity unified model may also reconstruct abnormal regions, resulting in a weak anomaly-score gap. Restoration-based methods alleviate this issue by introducing synthetic anomalies, but forcing the model to restore synthetic defects to normal appearances may overfit to the generated perturbations and does not necessarily generalize to real open-set defects.

We therefore use synthetic anomalies in a different way. Instead of regarding them as realistic defect templates or restoration targets, we treat them as class-agnostic counterexamples. Their role is to define what should be separated from the compact normal response space. This design is aligned with the regular industrial setting: it uses only normal real images, avoids category-specific assumptions, and encourages more stable anomaly-score separation for unified thresholding.

2.1.2. Architecture

Modified INP-Former backbone. We adopt INP-Former as the base architecture because it supports unified anomaly detection without relying on category-specific reference retrieval. Given an input image, a pretrained encoder extracts dense visual features. The INP-Former structure then derives intrinsic normal prototypes from the image itself and uses them to guide feature reconstruction. The discrepancy between the original features and the reconstructed normal features forms the reconstruction-based anomaly response.

Compared with the baseline implementation, we remove the gather loss and keep the training objective focused on normal reconstruction, abnormal repulsion, and residual localization. This modification simplifies the optimization and avoids introducing an additional prototype aggregation constraint that may conflict with the asymmetric separation objective.

Class-agnostic synthetic anomaly generation. During training, synthetic abnormal samples are generated by injecting random noise into local regions of defect-free images. The same generation rule is used for all categories. Let $\mathbf{x}$ denote a normal training image, and let $\mathbf{m} \in \{0,1\}^{H\times W}$ denote the generated binary anomaly mask. The perturbed image $\tilde{\mathbf{x}}$ is obtained by replacing or blending the masked region with random noise:

$$\tilde{\mathbf{x}} = (1-\mathbf{m}) \odot \mathbf{x} + \mathbf{m} \odot \mathbf{n}, \tag{1}$$

where $\mathbf{n}$ denotes the generated noise pattern. These synthetic regions are not intended to mimic the exact appearance of real defects. Instead, they provide abnormal spatial masks and counterexamples for learning normal–abnormal separability.

Asymmetric counterexample constraint. Let d_i denote the feature discrepancy at spatial location i. For normal locations Ω_n, the model minimizes the discrepancy and encourages a compact normal response:

$$\mathcal{L}_n = \frac{1}{|\Omega_n|} \sum_{i\in\Omega_n} d_i. \tag{2}$$

For synthetic abnormal locations Ω_a, we do not define a clean restoration target. Instead, abnormal regions are treated as counterexamples and are pushed beyond a normality margin:

$$\mathcal{L}_a = \mathbb{E}_{i\in\Omega_a} \left[\max(0, m - d_i)\right], \tag{3}$$

where m denotes the separation margin. The asymmetric counterexample constraint is then formulated as

$$\mathcal{L}_{ACC} = \mathcal{L}_n + \mathcal{L}_a. \tag{4}$$

This objective compacts normal responses while explicitly repelling synthetic abnormal regions from the normal response space, without forcing the model to restore synthetic anomalies to normal appearances.

Residual localization module. The asymmetric objective improves the separation between normal and abnormal responses, but feature discrepancy alone may not provide sufficiently precise pixel-level localization. To exploit the spatial prior provided by the synthetic anomaly mask, we introduce a lightweight residual localization branch.

Given the original feature $\mathbf{F}$ and the reconstructed normal feature $\hat{\mathbf{F}}$, we first compute the residual feature:

$$\mathbf{R} = \mathbf{F} - \hat{\mathbf{F}}. \tag{5}$$

The residual feature is then fed into a lightweight segmentation head $\Psi(\cdot)$ to obtain a dense residual anomaly map:

$$\mathbf{A}_{res} = \Psi(\mathbf{R}). \tag{6}$$

Different from restoration-based objectives, this branch does not require the model to recover clean normal pixels from corrupted regions. Instead, it directly learns a spatial anomaly response under the supervision of the synthetic anomaly mask. Let Ω_n and Ω_a denote the normal and synthetic abnormal pixel sets, respectively. The residual localization loss is defined as

$$\mathcal{L}_{RL} = \mathbb{E}_{i\in\Omega_n}\left[A_{res,i}^2\right] + \mathbb{E}_{j\in\Omega_a}\left[\left(A_{res,j}-1\right)^2\right]. \tag{7}$$

The first term suppresses residual responses in normal regions, while the second term encourages strong but bounded activations in synthetic abnormal regions. This direct pixel-level supervision helps the residual branch produce spatially aligned anomaly evidence and improves fine-grained localization.

Combining the asymmetric counterexample constraint with residual localization, the final training objective is

$$\mathcal{L} = \mathcal{L}_{ACC} + \mathcal{L}_{RL}. \tag{8}$$

2.1.3. Training

The model is trained under the regular setting using the official defect-free training images of MVTec AD 2. For each training image, pseudo-anomalous regions are generated online using the fixed random-noise strategy described above. The generated masks provide supervision for both the asymmetric counterexample objective and the residual localization branch.

All categories share the same model architecture, preprocessing pipeline, anomaly generation strategy, loss functions, and training schedule. No public-test annotations are used for training, validation, threshold selection, or hyperparameter tuning. All hyperparameters are fixed before final evaluation, ensuring that the method remains class-agnostic and compliant with the challenge setting.

2.2. Dataset & Evaluation

2.2.1. Benchmark dataset utilization.

We use the official MVTec AD 2 dataset from the VAND4.0 Industrial Track. Only defect-free images from the official training split are used as real training data. All images are uniformly resized to 1008×1008 before being fed into the model. During training, synthetic abnormal regions are generated online by a fixed class-agnostic random-noise perturbation strategy, and the corresponding masks supervise the asymmetric counterexample objective and the residual localization branch.

The same preprocessing, augmentation, synthetic anomaly generation, and inference pipeline are applied to all categories. The public test split is used only for sanity checking and qualitative analysis, without using its annotations for threshold selection or hyperparameter tuning. Final performance is evaluated by the official server on the hidden private splits.

2.2.2. Evaluation criteria.

The primary metric is SegF1, which measures pixel-level anomaly segmentation quality after binarizing the predicted anomaly map. It jointly reflects precision and recall, penalizing both false positives on normal regions and missed anomalous pixels.

For each test image, the model outputs a continuous anomaly map and a thresholded binary mask. The binary mask is obtained using a single global threshold estimated from the validation set. Specifically, we pool anomaly scores from all validation images across all categories and set the threshold to the 0.9995 quantile of the pooled score distribution. This threshold is fixed before final evaluation and is shared by all test images, categories, and splits.

2.2.3. Evaluation Setting

Our submission follows the regular setting of the Industrial Track. The model is trained on the official defect-free training images, together with automatically generated synthetic abnormal regions. The validation set is used only to estimate a global score threshold from its pooled anomaly-score distribution. No anomalous labels from the public or private test splits are used for training, validation, threshold selection, or hyperparameter tuning.

At inference time, the model takes only the test image as input and directly predicts the anomaly map. No test-time optimization, manual prompt, external reference retrieval, or synthetic anomaly generation is used during inference. The entire pipeline is class-agnostic: one shared architecture, one training strategy, one global thresholding rule, and one post-processing pipeline are used for all categories.

Table 1. Performance comparison on the MVTec AD 2 `test_public` split. Each entry denotes Pixel AUROC / SegF1.

Category	INP-Former	Ours
Can	0.848 / 0.019	0.844 / 0.001
Fabric	0.823 / 0.246	0.850 / 0.464
Fruit Jelly	0.961 / 0.428	0.969 / 0.533
Rice	0.965 / 0.390	0.976 / 0.694
Sheet Metal	0.868 / 0.210	0.914 / 0.393
Vial	0.922 / 0.612	0.919 / 0.356
Wallplugs	0.845 / 0.097	0.878 / 0.054
Walnuts	0.959 / 0.509	0.965 / 0.632
Average	0.899 / 0.315	0.914 / 0.391

3. Results

3.1. Performance Metrics

As shown in Table 1, our method improves the average Pixel AUROC from 0.899 to 0.914 and the average Pixel SegF1 from 0.315 to 0.391 on the MVTec AD 2 `test_public` split. The gain in Pixel AUROC indicates stronger anomaly-score discrimination, while the larger improvement in SegF1 suggests that our anomaly maps are more suitable for threshold-based pixel-level segmentation.

The improvement is especially evident on Fabric, Rice, Sheet Metal, and Walnuts, where SegF1 increases by a clear margin. These results show the effectiveness of random-noise counterexample learning and residual localization in producing more concentrated anomaly responses. Nevertheless, performance drops on Can, Vial, and Wallplugs, indicating that unified thresholding remains challenging under category-dependent score distributions. Overall, the results demonstrate that our method improves over the INP-Former baseline in both anomaly ranking and segmentation quality.

4. Discussion

4.1. Challenges & Solutions

The main challenge is learning anomaly segmentation from only defect-free images. Since real defects are unavailable and open-set, we use random-noise perturbations as class-agnostic counterexamples rather than realistic defect templates. The asymmetric counterexample constraint then encourages compact normal responses and pushes abnormal regions away from the normal response space.

Another challenge is the class-agnostic requirement. MVTec AD 2 contains diverse textures, structures, and defect scales, but category-specific models or thresholds are not used. Our method addresses this with a single shared architecture, unified training strategy, and global thresholding rule for all categories.

Threshold generalization remains difficult. Although Pixel AUROC measures score ranking, SegF1 depends on the thresholded mask. We mitigate this by improving normal–abnormal separability with asymmetric learning and residual localization, while using a validation-derived global threshold.

4.2. Model Robustness & Adaptability

The results show that our method improves average Pixel AUROC and SegF1 over the INP-Former baseline. The gains on Fabric, Rice, Sheet Metal, and Walnuts suggest that residual localization is effective for texture variations and fine-grained defects.

However, performance drops on Can, Vial, and Wallplugs indicate remaining limitations under reflective surfaces, transparent structures, and category-dependent score distributions. These cases show that robust score calibration and boundary localization are still critical for unified anomaly segmentation.

4.3. Future Work

First, better score calibration is needed to reduce the gap between continuous anomaly ranking and binary mask generation. Future work may explore stronger class-agnostic calibration strategies for categories with different score distributions and lighting conditions.

Second, pretrained segmentation models such as SAM may be used for automatic boundary refinement. Anomaly maps can provide spatial guidance, while the segmentation model refines coarse masks and improves boundary quality.

5. Conclusion

This report presents a unified unsupervised anomaly segmentation method for the VAND4.0 Industrial Track under the regular setting. Built on a modified INP-Former backbone, the method uses random-noise perturbations as class-agnostic counterexamples, asymmetric learning to improve normal–abnormal separation, and a residual localization branch to enhance pixel-level anomaly prediction. The results demonstrate the effectiveness of the proposed framework, while also highlighting score calibration and boundary refinement as important directions for future improvement.

References